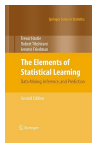


Lesson 02.6: ML Techniques - How to Do Research

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References:

- Burkov: *"The Hundred-Page Machine Learning Book"* (2019)
- Russell et al.: *"Artificial Intelligence: A Modern Approach"* (4th ed, 2020)
- Hastie et al.: *"The Elements of Statistical Learning"* (2nd ed, 2009)



- *How to Do Research*

Occam's Razor

- **Simple is better**
 - *"The simplest model that fits the data is also the most plausible"* (Occam)
 - *"An explanation of the data should be as simple as possible, but not simpler"* (Einstein?)
- Corollary:
 - Trim the model to the bare minimum necessary to explain the data
- What does **simple** mean?
 - A model is simple when it is one of few possible objects
- What does **better** mean?
 - A model is better means better out of sample performance
- **Why it's true?**
 - Less likely to fit a given data by coincidence
 - An event is more significant if it is unlikely (formalized in terms entropy)

What is a “simple model”?

- An object is **simple** when it is one of few possible objects
- **Examples:**
 - Polynomial of order 2 is simpler than order 17
 - * More polynomials of order 17 than order 2, though both are infinite sets
 - SVM (Support Vector Machine) characteristics:
 - * Separating hyperplane appears wiggly, but defined by few support vectors
- **Measures of complexity**
 - Complexity of a single hypothesis h
 - * E.g., polynomial order, MDL (describe hypothesis in bits), Kolmogorov complexity
 - Complexity of a hypothesis set $\mathcal{H}$
 - * E.g., VC dimension of the model
 - Complexity of h and $\mathcal{H}$ related by counting:
 - * If you need l bits to specify h , then h is one of 2^l elements of set $\mathcal{H}$

Model Soundness

- **Model should tell a story**
 - You cannot blindly accept results of modeling
 - Ask: *“What criticisms would you give if the model was presented for the first time?”*
- **Benchmark models**: compute performance if model outputs:
 - Always 0 or 1
 - * E.g., long-only model for stock predictions
 - Random results
 - * E.g., bootstrap of null hypothesis “no prediction power”
- **Perfect fit can mean nothing**, e.g.,
 - Get 2 data points on a plane
 - * Fit data with a linear relationship
 - * Perfect fit
 - Means nothing since:
 - * Always a line between 2 points
 - * Data cannot falsify hypothesis
 - The model (line) is too complex for data set (only 2 points)

Sampling Bias (1/2)

- **Sampling bias** occurs when data is not representative of the intended population
 - Biased sampling leads to biased outcomes
 - Model views world through training data
 - Hoeffding's hypothesis: training and testing distributions are the same
- **Causes of Sampling Bias**
 - *Non-random sampling*: Favors certain outcomes or groups
 - *Undercoverage*: Inadequate representation of some population members
 - *Survivorship bias*: Includes only successful subjects
 - *Self-selection bias*: Participants choose to participate, introducing bias
- **Consequences**
 - Systematic errors distort relationships or predictions
 - Models trained on biased samples perform poorly on real-world data
 - Leads to incorrect or unfair decisions

Sampling Bias (2/2)

- **Examples**

- Training facial recognition on lighter-skinned faces reduces accuracy on darker-skinned faces
- Predicting income using data only from employed individuals omits unemployed people

- **Mitigation**

- Compare sample statistics with population statistics
- Use stratified sampling or resampling to balance data
- Apply re-weighting or bias correction techniques
- If data points have zero probability ($\text{Pr} = 0$), no remedies possible

Data Snooping (1/2)

- **Data snooping** occurs when test set information improperly influences model-building
- **Typical Sources**
 - *Train-test contamination*: Test data leaks into training
 - * E.g., Using test data during feature engineering, model selection, or tuning
 - *Multiple hypothesis testing*: Trying many models inflates performance
- **Consequences**
 - Overly optimistic evaluations
 - Models perform well in validation but fail on unseen data
- **Why It's Dangerous**
 - Misleading results
 - False confidence in model quality
 - Common pitfall in machine learning workflows

Data Snooping (2/2)

- **Examples**

- Choosing features based on the full dataset before splitting gives foresight not present in deployment
- E.g., demean the entire data and then split in train / test

- **Preventive Strategies**

- Keep training, validation, and test sets separate
- Use cross-validation properly
- Fit preprocessing steps only on training data, then apply to test data

“Burning the Test Set”

- **Problem**

- Repeated test set use during development leaks into training
- Overfitting to test data
- *“If you torture the data long enough, it will confess whatever you want”* (Coase, 1982)

- **Symptoms**

- Test accuracy improves; real-world performance stagnates or degrades
- Model evaluation becomes untrustworthy

- **Solutions**

- *One-time use principle*: Evaluate on test set once, post-tuning
- *Use a validation set*: For model development and tuning
- *Statistical adjustments*:
 - * Adjust p-values for multiple testing
 - * Use Bonferroni or False Discovery Rate corrections
- *Theoretical tools*:
 - * VC dimension reflects learning capacity, including repeated trials
 - * Minimum Description Length (MDL) penalizes complex models and repeated fitting

How to Achieve Out-Of-Sample Fit

- **Goal:** Choose an hypothesis $g \in \mathcal{H}$ approximates the unknown target function f
 - What does $g \approx f$ mean?

$$g \approx f \iff E_{out}(g) \approx 0$$

- **Solution:**
 - Achieve:
 - * Good in-sample performance $E_{in}(g) \approx 0$
 - * Good generalization $E_{out}(g) \approx E_{in}(g)$
 - Together $\implies$ good out-of-sample performance $E_{out}(g) \approx 0$

What If Out-Of-Sample Fit Is Poor?

- The model performs well in sample ($E_{in} \approx 0$) but poorly out of sample ($E_{out} \gg E_{in}$)
- **What does it mean?**
 - In-sample performance is optimistic
 - Model is overfitted and fails to generalize
- **What do you do?**
 - Run diagnostics before long-term projects
 - Gain insight on what works/doesn't to improve performance
 - * E.g., bias-variance curves and learning curves
- **How to fix?**
 1. Training data
 - Get more training data $\iff$ fixes high variance
 2. Features
 - Remove features $\iff$ fixes high variance
 - Add features $\iff$ fixes high bias
 - Add derived features (e.g., polynomial features) $\iff$ fixes high bias

Why Using a Lot of Data?

- **Several studies** show that:
 - Different algorithms/models have remarkably similar performance
 - Increasing training set improves performance
- **Consequence**
 - High capacity model + massive training set = good performance
- **Why**
 - Using a high capacity model with many parameters (e.g., neural network)

$$E_{in} \approx 0$$

due to low bias (and high variance)

- A massive data set helps avoid overfitting

$$E_{out} \approx E_{in}$$

- These two conditions together

$$E_{out} \approx E_{in} \approx 0 \implies E_{out} \approx 0$$

What to Do When You Have Lots of Data?

- You have $m = 100M$ examples in data set: great!

- **Cons**

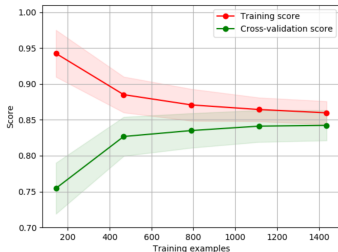
- Scalability issue (slow, lots of compute)
- Require work on infrastructure

- **First step:**

- Plot learning curves (in-sample and out-of-sample performance) for increasing amount of data
 $m = 1k, 10k, 100k, 1M, \dots$

- **Next step:**

- If model has large bias
 - * Training and validation performance are similar at $1M$
 - * Next step: use more complex model rather than training on $100M$ instances
- If model has large variance



Why We Do Things?

- **Always, always, always:**
 - Understand the purpose of the task
 - * Ask: *"Why are we doing something?"*
 - Clarify goals and outcomes of the task
 - * Ask: *"What do we hope to determine by performing the task?"*
 - Think about actions with the bigger picture in mind
 - Avoid going through the motions
 - Prioritize tasks by importance and impact
- E.g., when conducting a customer survey:
 - Q: *"Why is the feedback being collected?"*
 - * A: To improve product features and customer service
 - Q: *"What is the desired outcome?"*
 - * A: To identify areas for improvement and innovation
- E.g., before starting a marketing campaign:
 - Q: *"Why is this campaign run?"*
 - * A: To increase brand awareness or drive sales
 - Q: *"What are the specific goals?"*
 - * A: Set target number of new leads or click-through rates

Summary of the Results, Next Steps

- **Always have a summary of results**
 - High-level map of discoveries
 - * E.g., “smoothing model coefficients helps”
 - Highlight major findings
 - Interpret results
 - * E.g., *“Increase in sales likely due to new marketing strategy.”*
 - Conclusions
 - * Summarize data suggestions or confirmations
 - * E.g., *“Hypothesis that user engagement increases retention is supported”*
- **Always have a reference to detailed results**
 - Provide quick insights before details
- **Always have next steps**
 - What do you expect to happen?
 - * Like thinking n moves ahead in chess
 - E.g., *“Next, conduct detailed analysis on demographics contributing most to sales growth”*
 - * Outline potential experiments or analyses to validate findings further

Example: Spam Filter Classification

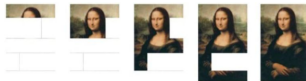
- You use $N = 5$ words in an email to distinguish spam / non-spam emails using logistic regression
 - Words are `buy`, `now`, `deal`, `discount`, `<your name>`
- **Goal:** improve classifier performance
 1. **Collect more data**
 - Honeypot project: fake email account to collect spam
 2. **Use better features**
 - Email routing info (spammers use unusual accounts, mask emails as legitimate)
 3. **Use better features from message body**
 4. **Detect intentional misspellings**
 - Spammers use misspelled words (e.g., `w4tch` for `watch`)
 - Use stemming software

Right and Wrong Approach to Research

- It is not clear how to prioritize the different possible tasks
- **Bad**
 1. Use gut feeling to choose a task
 2. Complete task
 3. Re-evaluate performance
- **Good**
 1. Build a simple algorithm
 - Within 1 day
 2. Set up performance evaluation
 - Single number and bounds to improve
 - Use cross-validation
 3. Set up diagnostic tools
 - Compute learning and bias-variance curves
 - Understand issues before fixing (avoid premature optimization)
 4. Review misclassified emails manually
 - Identify features to improve performance
 - E.g., what are types of misclassified emails?

Incremental vs Iterative

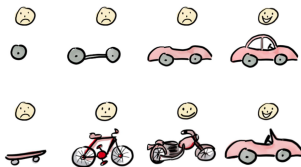
- **Incremental Development**
 - Each increment adds functional components
 - Require upfront planning to divide features meaningfully
 - Integration of increments can be complex
- **Iterative Development**
 - Each increment delivers usable system
 - Refine and improve product through repeated cycles
 - Get feedback
 - Uncover and adjust for unknown requirements
- **Incremental** >> **Iterative**



_Incremental



Iterative



Incremental vs Iterative